Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Technische Universität Dresden
Short name / acronym used in projects	TUD
Participant Identification Code (PIC), if available	999897729
Website(s)	https://tu-dresden.de/ Marius Ader / Ader Lab: https://tu-dresden.de/cmcb/crtd/forschungsgruppen/crtd-forschungsgruppen/ader Additional research-group page: https://www.digs-ils.phd/research/research-groups/marius-ader
Country	Germany
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Fetscherstr. 105, 01307 Dresden, Germany
Department(s), institute(s), research group(s) or unit(s) that would be involved	Center for Regenerative Therapies Dresden (CRTD), Center for Molecular and Cellular Bioengineering (CMCB), Technische Universität Dresden. Marius Ader group / Cell Replacement in the Mammalian Retina.
Additional institutional resources relevant to collaboration	CRTD/CMCB regenerative-medicine environment; retinal organoid and pluripotent-stem-cell-based research; preclinical retinal degeneration models; broader Dresden expertise in regenerative biology, organoids and model organisms.

Main contact person for this profile	Prof. Dr. Marius Ader, PI / Full Professor, CRTD/CMCB, Technische Universität Dresden
Main contact email	marius.ader@tu-dresden.de
Other contact person(s), if relevant	Jochen Hentschel, Lab manager, jochen.hentschel@tu-dresden.de

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☒ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☒ Advanced therapies / regenerative medicine	☒ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	retinal organoids; iPSC; photoreceptor transplantation; RPE replacement; cell replacement; retinal degeneration; photoreceptor integration; retinal regeneration; preclinical models; cell therapy
Geographies of interest for exchanges (optional)	Any

Overview of your research / innovation focus
The Ader Lab at the Center for Regenerative Therapies Dresden focuses on the development of cell-based strategies for the treatment of retinal degeneration, with particular emphasis on replacing lost photoreceptors and retinal pigment epithelial (RPE) cells. The group generates photoreceptor-containing retinal organoids and RPE from pluripotent stem cells, including human induced pluripotent stem cells (iPSCs), and evaluates their potential in preclinical models of retinal degeneration. Research addresses the isolation and enrichment of rod and cone photoreceptors and RPE, transplantation into degenerating retina, and analysis of donor-cell morphology, maturation, synaptic connectivity and functional responses. A major research interest is understanding the cellular and molecular requirements for successful donor-cell interaction with the host retina. The group's work includes studies of photoreceptor material exchange, donor-cell integration and functional repair. Current directions include defining conditions that optimise photoreceptor and RPE replacement and exploring sequential co-transplantation approaches. The overall orientation is preclinical and translational, with the long-term aim of developing cell-based therapeutic strategies for blinding retinal diseases.

Why your team could be relevant to a translational eye-health and vision consortium
The Ader Lab could provide a complementary regenerative-retina and cell-replacement capability within an IOBA-led translational eye-health network. Its expertise in human iPSC-derived retinal organoids, photoreceptor and RPE preparation, transplantation and functional analysis is particularly relevant to consortium activities addressing inherited

and degenerative retinal disease. The group offers an important bridge between human cell-based models and in vivo preclinical validation, allowing disease mechanisms and therapeutic concepts to be assessed in complex retinal systems. The strongest potential complementarity identified with IOBA concerns retinal disease modelling and regenerative therapy: IOBA's work on PROM1-associated retinal degeneration, patient-derived iPSCs, retinal organoids and neuroretinal models could be connected with Dresden expertise in photoreceptor/RPE replacement, transplantation and functional validation. This could create opportunities for knowledge transfer in both directions, from disease modelling and clinically informed research questions towards advanced regenerative validation, and from transplantation and cell-integration expertise towards improved preclinical therapeutic development.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Generation and use of retinal organoids and RPE from pluripotent stem cells, including human iPSC-derived material.
- Isolation and enrichment of rod and cone photoreceptors and RPE from stem-cell-derived retinal organoids.
- Retinal transplantation approaches for photoreceptor and RPE replacement in preclinical retinal degeneration models.
- Preclinical mouse models of retinal degeneration and assessment of donor-cell behaviour in degenerating retina.
- Analysis of transplanted donor cells using morphology, immunohistochemistry/immunocytochemistry, expression patterns and assessment of synaptic connections.
- Functional assessment of retinal repair, including light-response measurements and electrophysiological approaches such as electroretinography.
- Investigation of donor-cell integration, maturation and interaction with host retinal cells, including Müller glia and second-order neurons.
- Research into cytoplasmic/material transfer between photoreceptors and its implications for retinal cell-replacement strategies.
- Development of strategies for combined photoreceptor and RPE replacement, including sequential co-transplantation.
- Knowledge transfer on regenerative-retina models, transplantation workflows and preclinical validation of cell-replacement concepts.

Infrastructure, equipment, facilities or datasets relevant to consortium work

- CRTD/CMCB research environment focused on regenerative medicine and cell-based approaches.
- Cell culture and pluripotent-stem-cell-derived retinal organoid/RPE research capacity.
- Preclinical retinal degeneration models, particularly mouse models used for photoreceptor/RPE transplantation studies.
- Platforms and expertise for retinal transplantation and post-transplant donor-cell analysis.
- Immunohistochemical/immunocytochemical and microscopy-based analysis of retinal tissue and transplanted cells.
- Functional retinal assessment, including electroretinography and light-response analysis.
- Broader Dresden ecosystem of regenerative biology, organoid and model-organism expertise, including additional potential complementarities identified during the meeting.
- A range of core facilities including light microscopy, electron microscopy, mouse house, genome center, flow cytometry and stem cell engineering are available in house.

Training capacity and knowledge-transfer potential

- **Hands-on training in retinal cell-replacement research**, from the generation and characterisation of iPSC-derived retinal organoids to the preparation of photoreceptor and RPE populations for transplantation studies.
- **Practical training in preclinical retinal transplantation models**, including experimental design, transplantation procedures and evaluation of donor-cell behaviour in models of retinal degeneration.

- **Training in the assessment of retinal repair**, combining structural and cellular analyses with evaluation of donor-cell maturation, interaction with host retinal cells, synaptic connectivity and functional responses.
- **Exposure to translational development of cell-based therapies**, providing insight into how experimental findings on photoreceptor and RPE replacement can inform the development of therapeutic strategies for retinal degeneration.

What your team would hope to gain from participating
• Access to clinically informed retinal disease models and translational research questions, particularly inherited retinal degeneration and patient-derived iPSC systems. • Exchange with teams working on PROM1 and other genetically defined retinal diseases, retinal organoids and disease modelling. • Exposure to complementary neuroprotection, retinal explant, gene-editing and clinically oriented validation approaches. • Development of longer-term collaboration between Dresden regenerative-retina research and IOBA's translational ophthalmology environment. • Joint research and knowledge-transfer opportunities linking cell replacement, disease modelling and functional retinal validation.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Prof. Dr. Marius Ader	PI / Full Professor	Center for Regenerative Therapies Dresden (CRTD), an institute of the Center for Molecular and Cellular Bioengineering (CMCB); Ader Lab	Research staff / PI	Experienced Researcher	marius.ader@tu-dresden.de

Main disciplines represented in your team	Retinal biology; regenerative medicine; cell replacement; stem-cell biology; retinal organoids; photoreceptor biology; RPE biology; transplantation; preclinical retinal disease models; microscopy/immunohistochemistry; electrophysiology, vision-guided behavioral test (Light/dark box, opto-kinetic tracking)
Experience in EU / international / multi-partner collaborative projects	Marius indicated that he had not been very active in EU projects previously but is becoming more involved. He expressed interest in building international collaboration through MSCA Staff Exchanges and recommended the COST Action 'Retina 4 Future' as a networking route.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☒ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
• **Tessmer K, Hammer J, Gasparini SJ, Schmidtke K, Adhikari T, Michalke L, Loureira LR, Kurtz A, Last A, Chelius N, Johannsen H, Carido M, Ferguson S, Petzold A, Friedrich UA, Kurth T, Dahl A, Feldmann A, Almedawar S, Knöbel S, Ader M (2026).** Extensive monolayer formation after transplantation depends on a subpopulation of human iPSC-derived RPE cells. **iScience, accepted.**

- **Seidemann, S., Salomon, F., Hoffmann, K. B., Kurth, T., Sbalzarini, I. F., Haase, R. & Ader, M. (2024).** Automated quantification of photoreceptor outer segments in developing and degenerating retinas on microscopy images across scales. **Frontiers in Molecular Neuroscience, 17, 1398447.** DOI: **10.3389/fnmol.2024.1398447**

- **Carido, M., Völkner, M., Steinheuer, L. M., et al., including Ader, M. (2023).** *Reliability of human retina organoid generation from hiPSC-derived neuroepithelial cysts.* **Frontiers in Cellular Neuroscience, 17, 1166641.** DOI: **10.3389/fncel.2023.1166641**

- **Gasparini, S. J., Tessmer, K., Reh, M., et al., including Ader, M. (2022).** *Transplanted human cones incorporate into the retina and function in a murine cone degeneration model.* **Journal of Clinical Investigation, 132(12), e154619.** DOI: **10.1172/JCI154619**

- **Heisterkamp, P., Borsch, O., Diaz Lezama, N., et al., including Ader, M. (2022).** Evidence for endogenous exchange of cytoplasmic material between a subset of cone and rod photoreceptors within the adult mammalian retina via direct cell-cell connections. **Experimental Eye Research, 219, 109033.** DOI: **10.1016/j.exer.2022.109033**

Up to 5 relevant previous projects, grants, networks or collaborations
Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***   • COST Action 'Retina4Future' – CA24105. Partner. (Oct 2025 – Oct 2029) (https://retina4future.eu/) • BMBF / ERA Net Neuron 01EW2106, Human organoid system based therapy development for age-related macular degeneration. Beneficiary. (2021 – 2025) • DFG SPP2127 Gene and cell based therapies to counteract neuroretinal degeneration (2018 – 2025). Beneficiary. https://www.spp2127.de • BMBF ReSight. Coordinator. (2017 – 2022). https://www.gesundheitsforschung-bmftr.de/de/resight.php.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
N/A

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☒ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☒ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☒ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	licenses for animal work necessary when doing in vivo work

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. 1. Human iPSC-derived retinal organoids and RPE for regenerative-retina research. 2. Photoreceptor/RPE transplantation and preclinical retinal degeneration models, including functional validation. 3. Expertise in donor-cell maturation, integration/interaction with host retina and cell-replacement mechanisms.

Capabilities or expertise you would especially like to access through the consortium
1. Patient-derived retinal disease models and clinically informed disease phenotyping. 2. Complementary gene-editing, neuroprotection, retinal explant and translational ophthalmology approaches. 3. Clinical validation pathways and potentially industrial/regulatory expertise relevant to cell-based retinal therapies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☒ Yes ☐ No
☒ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. NDA, institutional approval, ethics needed

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☒ **Understood**

Name: Marius Ader
Date completed: 04/10/2026

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